Concentration of the Australian Equity Market in 5 Major Banks

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Motivation

Australia's equity market exhibits significant concentration in the banking sector, where a few stocks account for a large share of the stock market capitalization

World's most expensive bank by P/E and P/B ratio

ASX: CBA	ASX: NAB	ASX: WBC	ASX: ANZ	ASX: MQG
~11% of ASX200	*~5% of ASX200*	*~5% of ASX200*	*~4% of ASX200*	*~3% of ASX200*

Problem

Index Concentration

The five major banks make up between 25% to 30% of the ASX200 index in the past 10 years

(*S&P Global, 2025*)

Market Distortion

If share price dependence between the banks is strong, it disproportionately drives ASX200 returns, volatility, and investor sentiment. Thus, it also reduces diversification benefit of investing into the ASX200 index directly

Portfolio Exposure

Superannuation funds and ETFs may include ASX200, concentrating household wealth in these five banks

Research Question

This presentation will investigate the following question using Mutual and Conditional Mutual Information

Research Question

To what extent do dependencies exist among the stock price movements of Australia's major banks, and do these dependencies persist after conditioning on broader stock market and macroeconomic influences?

Information Theoretic Tools

Take statistical POV
i.e., "Non-linear correlation"

Mutual Information

Conditional Mutual Information

Interpretation

Low MI, Low CMI → **Best Case**
Independent price dynamics

High MI, Low CMI → **Moderate Case**
Market/macro related interdependence (Systematic Risk)

High MI, High CMI → **Worst Case**
Banking sector-specific interdependencies (Idiosyncratic Risk)

Method

To address the research question, mutual and conditional mutual information are calculated over a 10-year period using the KSG1 estimator on 150-day rolling windows, capturing temporal dependencies among the banks' daily returns

10+ years historical daily prices retrieved via YFinance Python API

Daily log returns calculated and standardised; ACF performed

150-day rolling window applied to capture time-varying dependencies

MI and CMI calculated between all bank pairs using the KSG1 estimator

Additional Notes

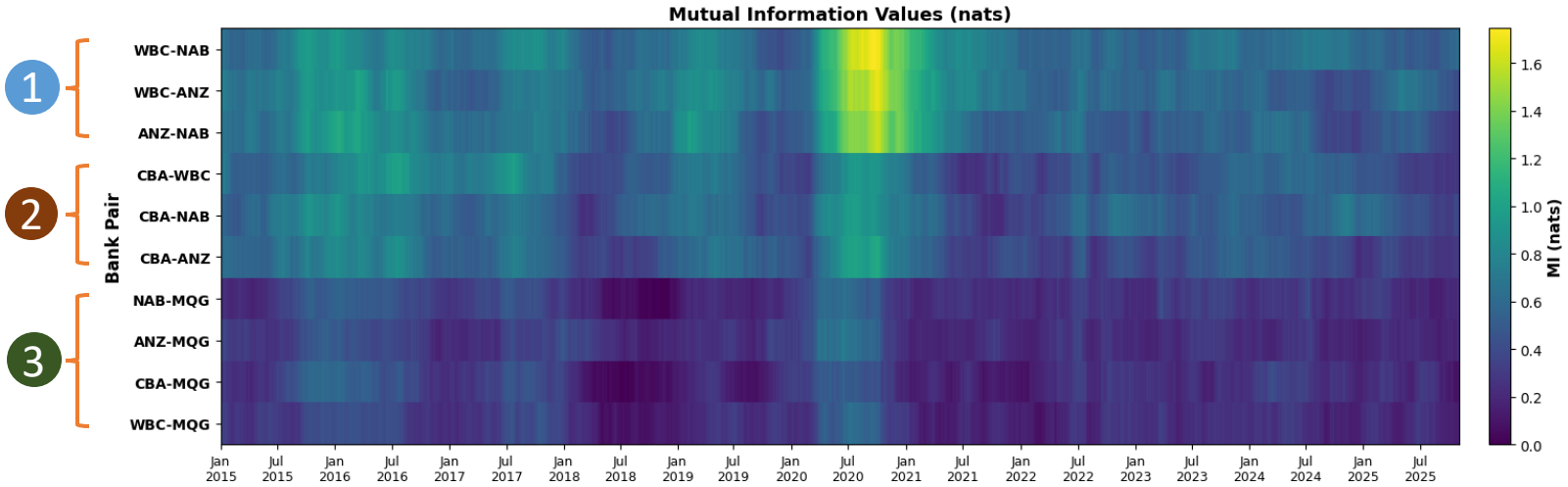
- **Robustness Checks:** After testing various values, window size set to 150 days, K-Neighbours set to k = 4, and selected KSG1 Algorithm for best bias-variance trade-off
- **Estimator Choice:** KSG preferred over Gaussian to capture non-linear relationship, and stock returns typically have fatter tails than the Gaussian Distribution
- **Autocorrelation Analysis:** ACF analysis used to set the Theiler window to avoid same-sample bias; generally insignificant autocorrelation so DYN_CORR_EXCL = 0

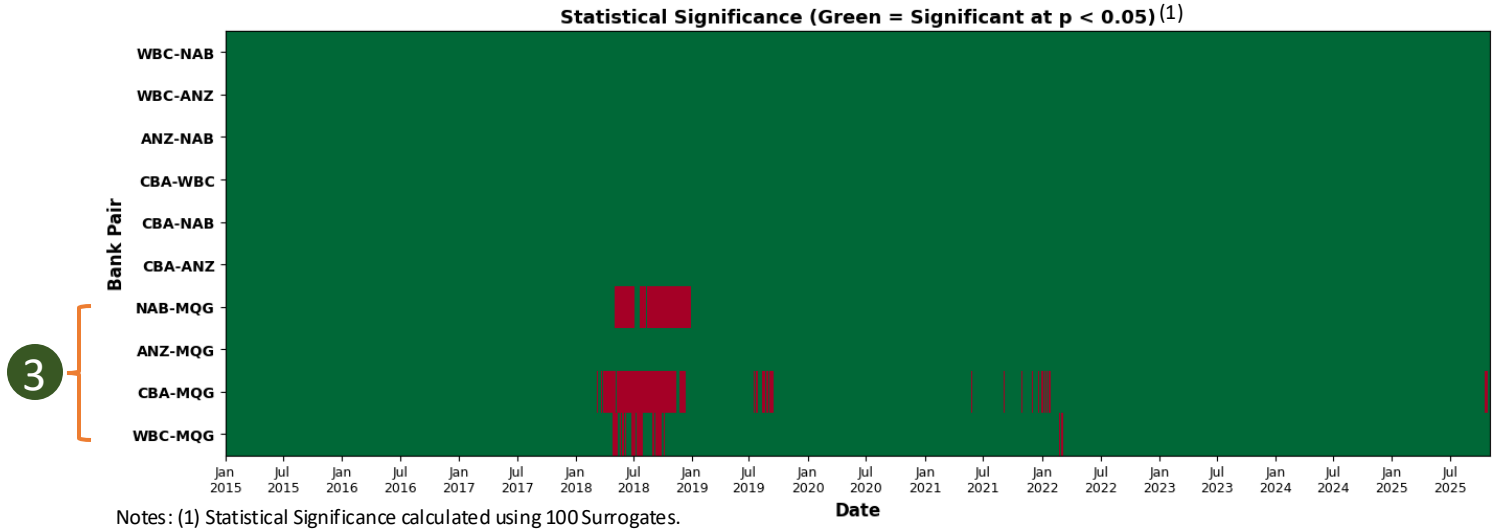
Conditioning Variables

Factor	Proxy	Rationale
Broader Stock Market Movement	Solactive ASX200 Ex-Top 20 Index (*DE000SL0CK81.SG*)	Controls for overall market trends while excluding the major banks. ASX200 Ex-Top 20 best available proxy w/o paying for financial data
Macroeconomic Factors	Solactive Australia Government Bonds Index (*DE000SLA6QZ3.SG*)	Bond yields reflects expectations of interest rates, inflation, and other macro factors. Bond yields update continuously, while underlying fundamentals update periodically

Mutual Information (Cross Sectional Analysis)

Strong dependencies exists among traditional retail banks, while CBA and Macquarie exhibit distinctive dynamics reflecting investor confidence and business diversification, respectively





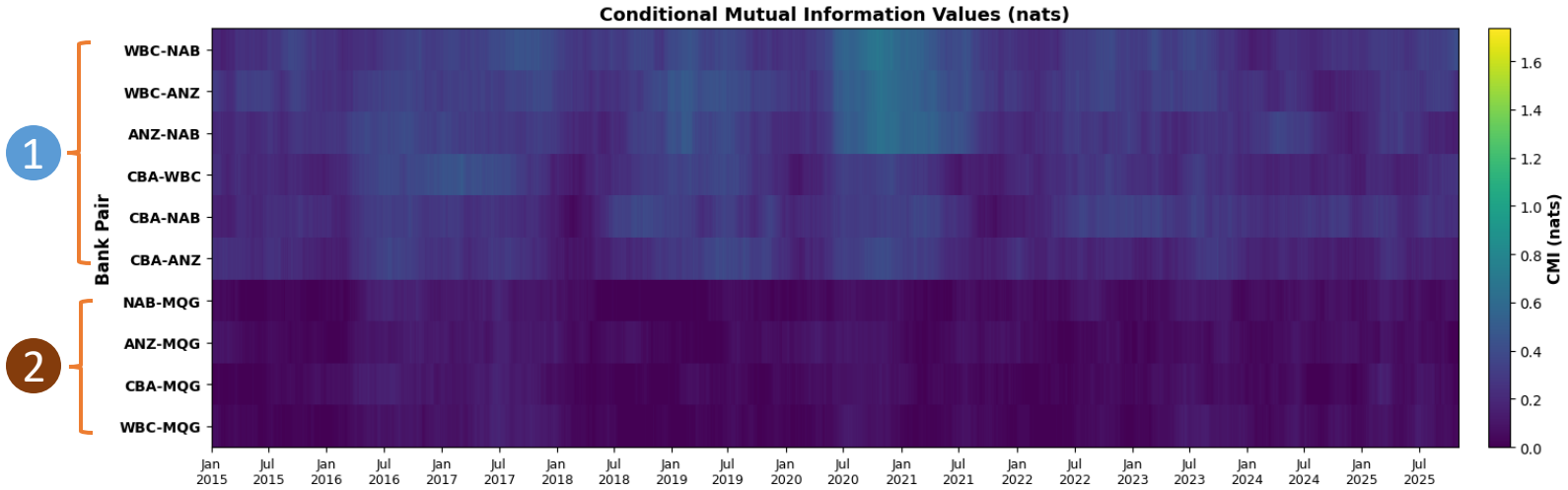
Notes: (1) Statistical Significance calculated using 100 Surrogates.

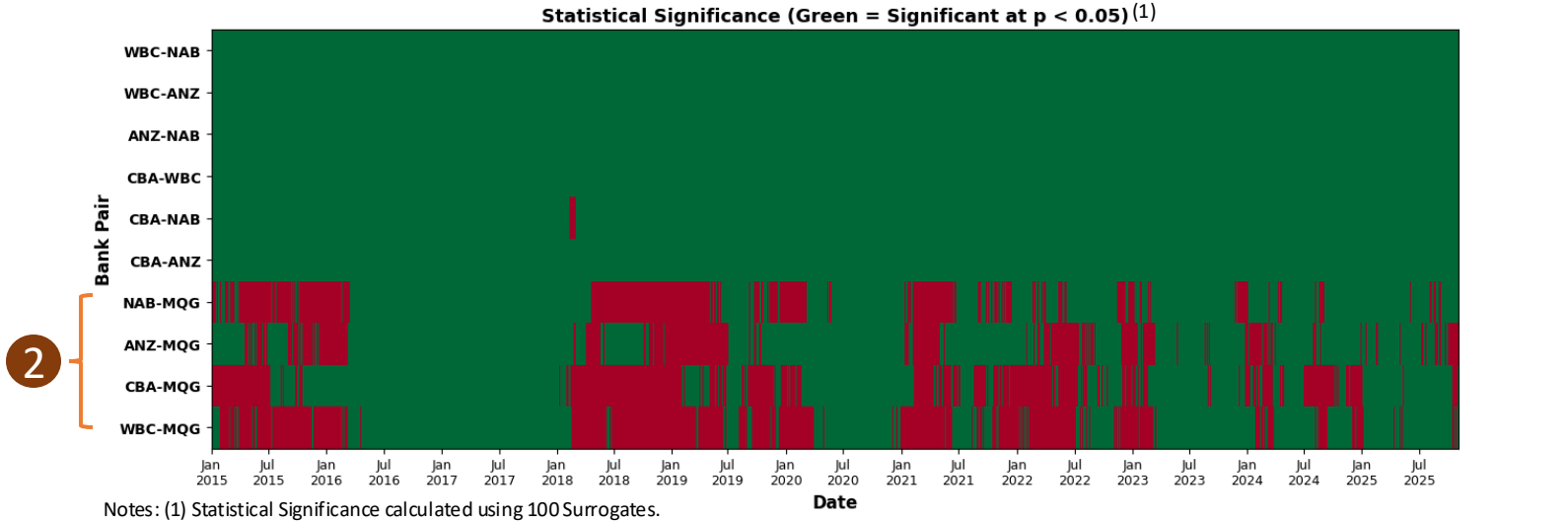
Key Findings

1. **Within-group pairs among Westpac, NAB, and ANZ exhibit the highest mutual information**, indicating strong stock price dependencies and likely structural similarity among these three retail banks
2. **Commonwealth Bank (CBA) pairs show comparatively weaker dependence**, likely because CBA is viewed by investors as a safer, more stable institution (largest ASX-listed stock, dominant market share, and benefits from higher investor confidence)
3. **Macquarie Group (MQG) pairs displays the weakest and least consistent mutual information**, with a few statistically insignificant periods, reflecting its more diversified business mix across asset management, infrastructure, and investment banking

Conditional Mutual Information (Cross Sectional Analysis)

After controlling for market and macroeconomic factors, the remaining dependencies among banks are substantially reduced, suggesting that much of the observed price dependence is explained by common external factors





Notes: (1) Statistical Significance calculated using 100 Surrogates.

Key Findings

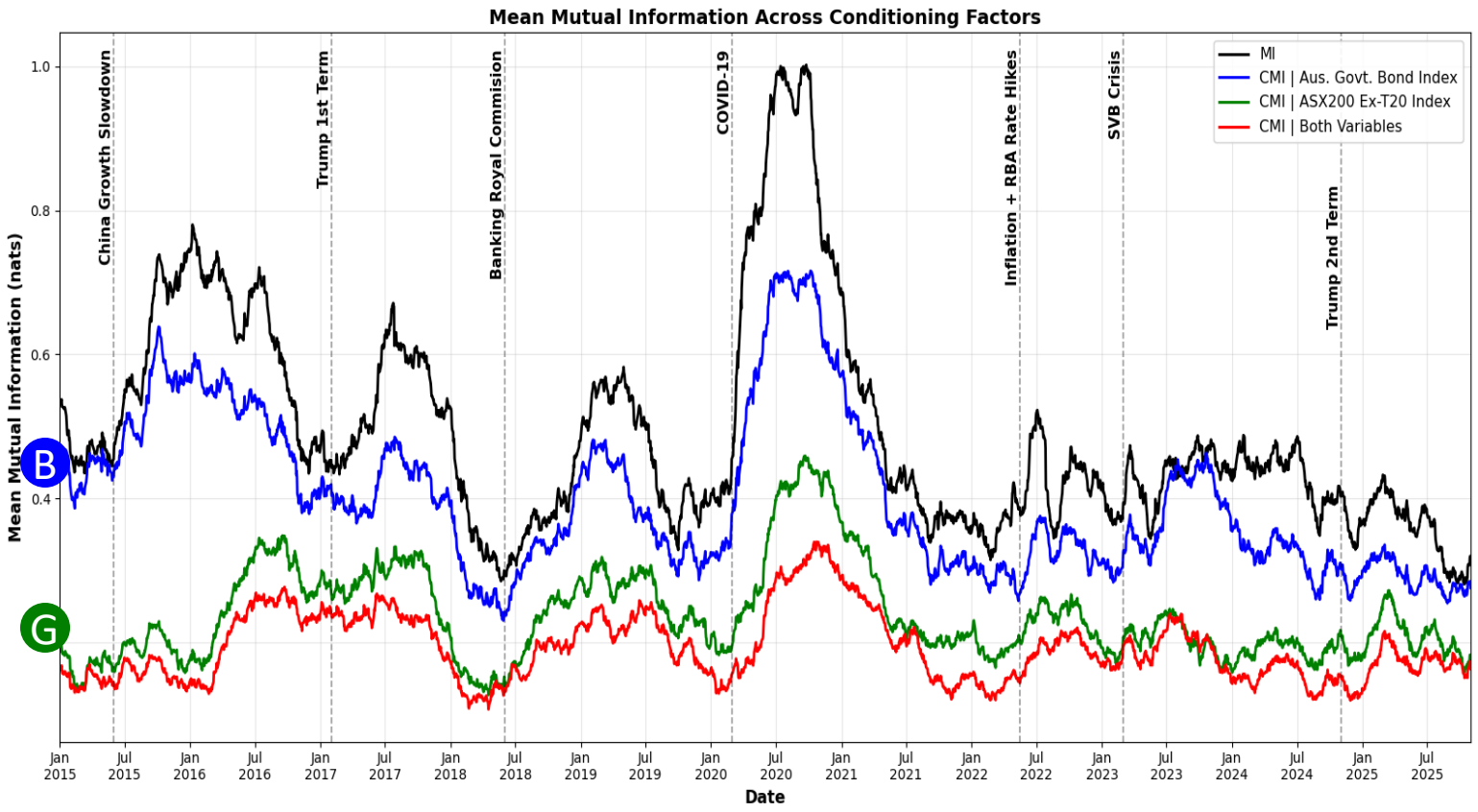
~ **Conditional mutual information values drop substantially** once controlling for broader market movements and macroeconomic factors, suggesting that much of the dependence between banks is explained by common market and macro factors

1 **Residual dependencies among the big 4 retail banks remain**, suggesting sector-specific linkages in stock price returns beyond broader market influences

2 **CMI values involving Macquarie (MQG) are now commonly statistically insignificant**, suggesting that its price dependencies with other banks is largely market/macro related rather than sector-influenced, consistent with its diversified business model

Additional Insights (Time Series Analysis)

Time-series analysis reveals that bank dependencies fluctuate with market conditions, intensifying during periods of stress. However, these dependencies weaken once market and macroeconomic factors are controlled



Key Findings

- ~ **Mutual information evolves over time, with pronounced peaks during systemic stress events** (e.g. COVID-19, RBA rate hikes), reflecting transient surges in interbank share price dependencies
- G **Conditioning on broader stock market movements (ASX200 Ex-Top 20) significantly dampens these peaks**, showing that much of the share price dependencies is explained by overall stock market movements
- B **Conditioning on macro factors (Australian Government Bond Yields) results in a smaller but time-varying reduction in dependence**, suggesting sensitivity to changing macroeconomic and policy environments.

Conclusion

Share price dependence among the major Australian banks are mostly market and macro related

Conclusion

Low MI, Low CMI	**Best Case** Independent price dynamics
High MI, Low CMI	**Moderate Case** Market/macro related interdependence (Systematic Risk)
High MI, High CMI	**Worst Case** Sector-specific interdependencies (Idiosyncratic Risk)

Final Takeaway

As shown by high MI and low CMI, share price dependence among Australia's major banks is **primarily related to market and macroeconomic forces**

Limitations / Future Work

- **Temporal Resolution:** Analysis is limited to daily data, which may overlook short-term relationships between bank prices. Future work could apply intra-day or higher-frequency data to capture faster market responses
- **Proxy Selection:** Conditioning only on two broad proxies (stock index and bond yields) which may simplify macro complexity. Expanding to additional indicators (e.g. credit spreads, funding costs, housing data) could better represent underlying economic drivers
- **Future Directions:** Extend analysis using more complex tools (e.g. transfer entropy or effective connectivity graphs) to map directional dependencies and potential information flow among banks and macro variables

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